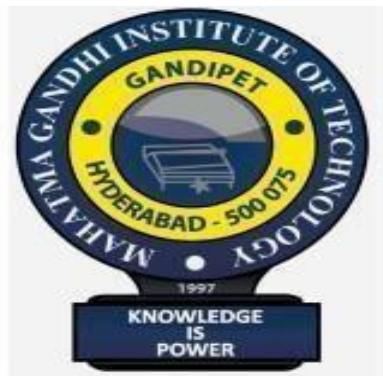
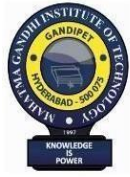


Data Visualization using Filters
A Project Report
Submitted in partial fulfilment for the award of the degree
Of
Bachelor of Technology
In
Computer Science and Engineering
By
K. Navadeep (23261A05G1)
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Under the guidance of
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YEARS

We **K. Navadeep** and **P. Rishi Reddy**, bearing the roll numbers, **23261A05G1** and **23261A05H7**, hereby declare that the project entitled Data Visualization using Filters under the guidance of **Dr K. Rajitha, ASSISTANT PROFESSOR**, Department of Computer Science and Engineering, MGIT, Hyderabad, is submitted in the partial fulfilment of the requirement for qualifying the project work course during the academic year 2024-2025.

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ABSTRACT

This research paper introduces data visualization tools and techniques in various domains. The importance of data visualization in enhancing the understanding and communication of complex data is discussed, along with the various types of data visualization tools and techniques available.

The paper also covers the use of data visualization in business, social sciences, humanities, sports, environmental sciences, and healthcare. The paper concludes with a summary of the key takeaways and the potential impact of data visualization tools and techniques in various domains.

The paper aims to provide a comprehensive overview of data visualization tools and techniques and to highlight their importance in various fields for effective data communication and various analysis for the better examination of larger data.

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INTRODUCTION

Data visualization is an essential tool for understanding complex data and communicating insights effectively. Tableau is one of the most widely used platforms for data visualization, offering powerful and interactive features to help users turn raw data into meaningful visual stories.

Whether you're analysing business performance, exploring trends, or identifying patterns, Tableau provides a wide range of features to create visually appealing and insightful dashboards.

What is Tableau?

Tableau is a data visualization software that enables users to create a variety of interactive and dynamic charts, graphs, maps, and dashboards. It is known for its ease of use, drag-and-drop interface, and the ability to connect to a variety of data sources like Excel, SQL databases, cloud platforms, and even web-based data sources like Google Analytics.

Basics to know about Data Visualization

What is Data?

Data refers to raw facts, statistics, or information collected or stored in a structured or unstructured form. Data can take various forms, such as text, numbers, images, videos, and more. It is the foundation of all information and knowledge and is used in various fields for analysis, decision-making, and understanding trends and patterns.

Data can be categorized into two main types:

- **Structured Data:** This type of data is organized into a specific format, such as tables or databases, and is easily searchable and analyzable. Examples include spreadsheets, relational databases, and CSV files.
- **Unstructured Data:** Unstructured data lacks a specific format and can include text documents, social media posts, images, audio recordings, and more. Analyzing unstructured data often requires advanced techniques like natural language processing and image recognition.

Where to Find Data?

You can find data from various sources, depending on your specific needs:

- **Open Data Portals:** Many governments and organizations provide free access to a wide range of data through open data portals. Examples include Data.gov (United States) and data.gov.uk (United Kingdom).
- **Data Repositories:** Academic institutions, research organizations, and data enthusiasts often share datasets on platforms like Kaggle, GitHub, and the UCI Machine Learning Repository.
- **APIs (Application Programming Interfaces):** Some websites and services offer APIs that allow you to programmatically access and retrieve data. Examples include Twitter API, Google Maps API, and financial market APIs.
- **Web Scraping:** You can extract data from websites using web scraping tools and libraries like BeautifulSoup and Scrapy. However, be mindful of the website's terms of use and legal restrictions.
- **Surveys and Interviews:** You can conduct your own surveys or collect data through questionnaires and interviews.
- **IoT Devices:** Internet of Things (IoT) devices generate vast amounts of data that can be used for various purposes.

Foundations for Building Data Visualizations:

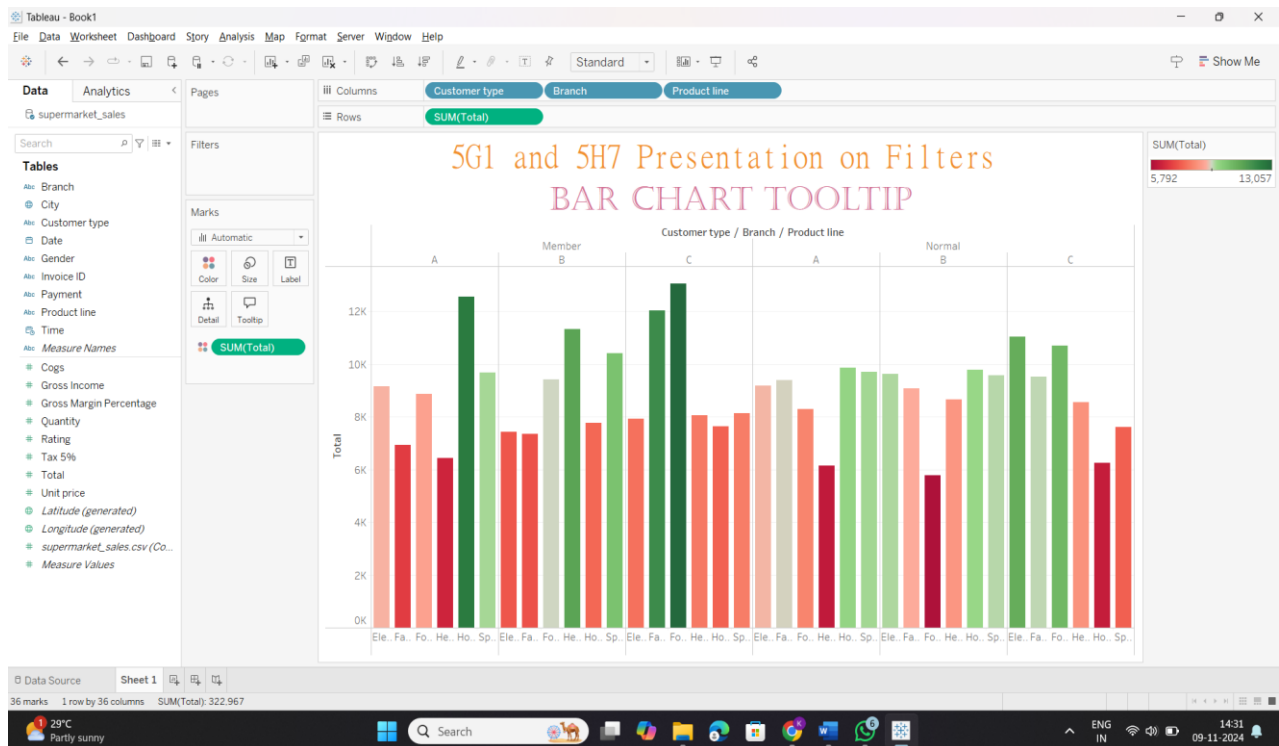
Creating effective data visualizations requires a strong foundation in several key areas:

- **Data Analysis:** Before creating visualizations, you should thoroughly analyze your data to understand its structure, relationships, and any patterns or trends. Exploratory data analysis (EDA) techniques can help with this.
- **Statistical Knowledge:** Understanding basic statistics is essential for making meaningful interpretations of data. Concepts like mean, median, standard deviation, and correlation are commonly used in data visualization.
- **Domain Knowledge:** Having knowledge of the specific domain or subject matter related to your data is crucial for creating contextually relevant visualizations. It helps you ask the right questions and provide valuable insights.
- **Visualization Tools:** Familiarize yourself with data visualization tools and libraries such as matplotlib, Seaborn, ggplot2, D3.js, and Tableau. Each tool has its strengths and can be used for different types of visualizations.

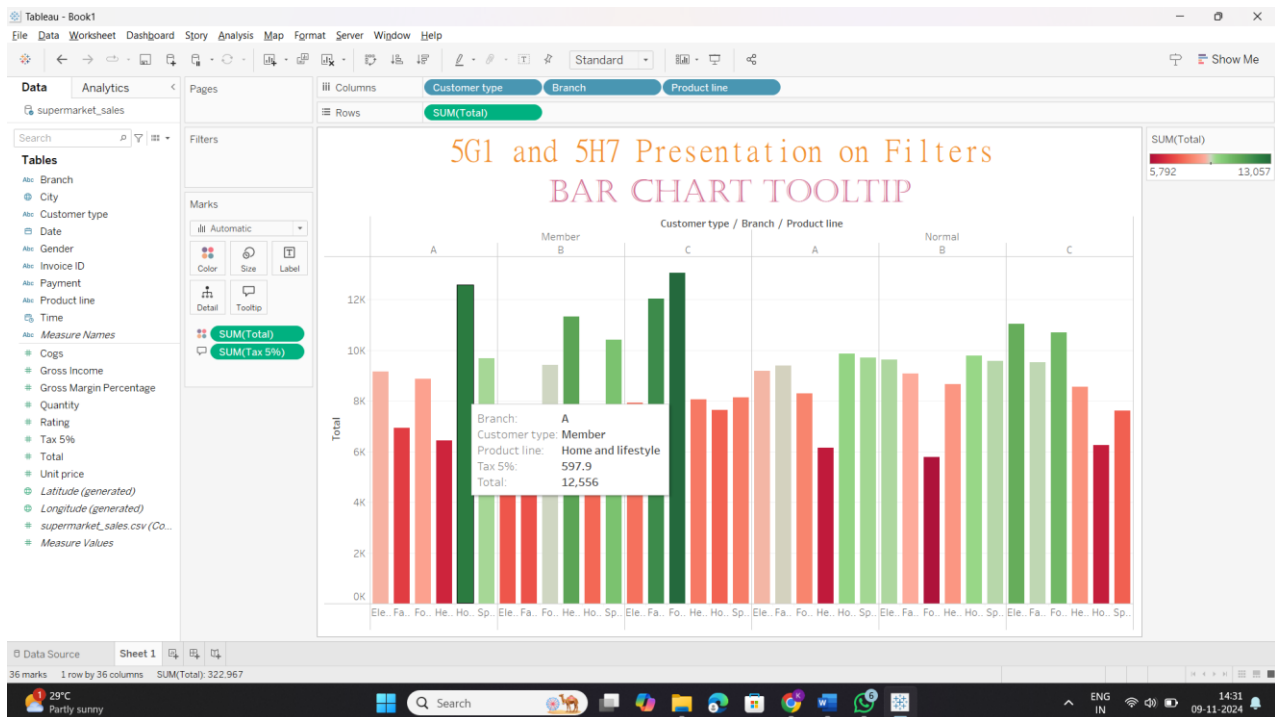
DATA VISUALIZATION USING FILTERS

Aim: Advanced Visualization Tools: Using Filters, Using the Detail panel, using the Size panels, customizing filters, Using and Customizing tooltips, formatting your data with colors.

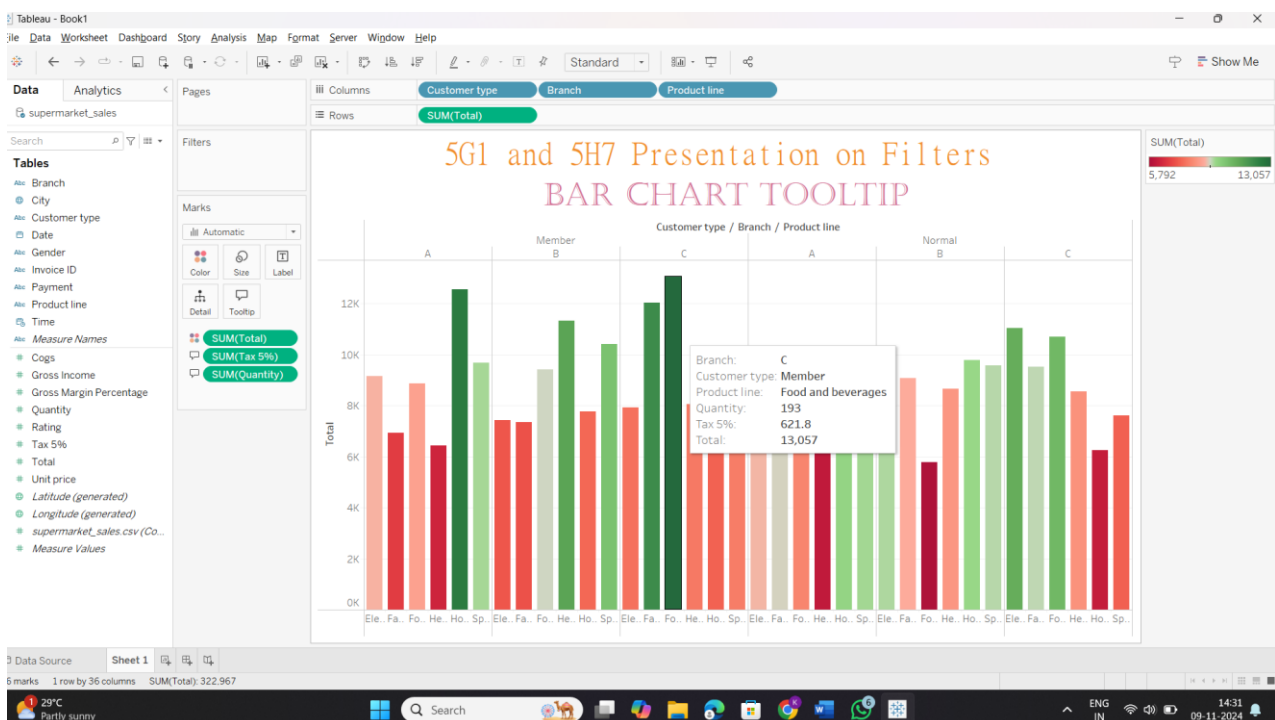
Step 01: Drag and Drop Customer type and Total in row and columns silts.



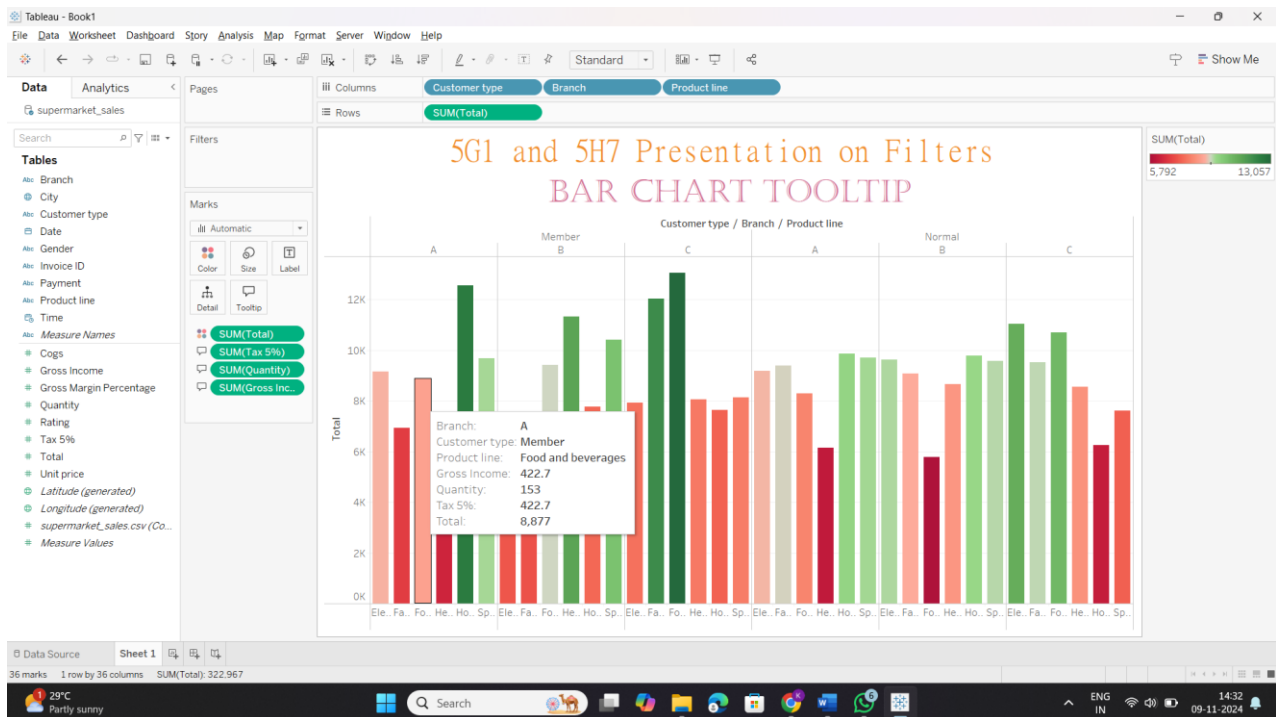
Step 02: Drag and drop the Tax field into tooltip marks card and after click any bar, automatically showing the profit in pop-up window.



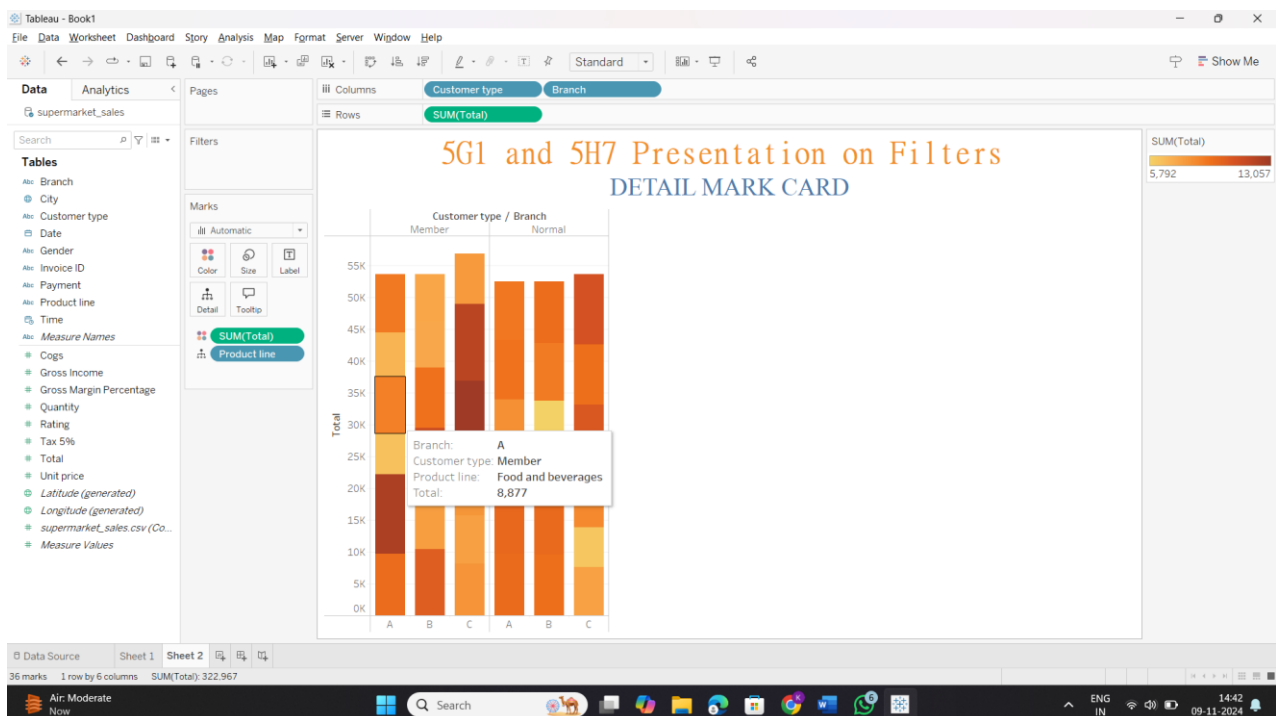
Step 03: Drag and drop the Quantity field into tooltip marks card and after click any bar, automatically showing the profit in pop-up window.



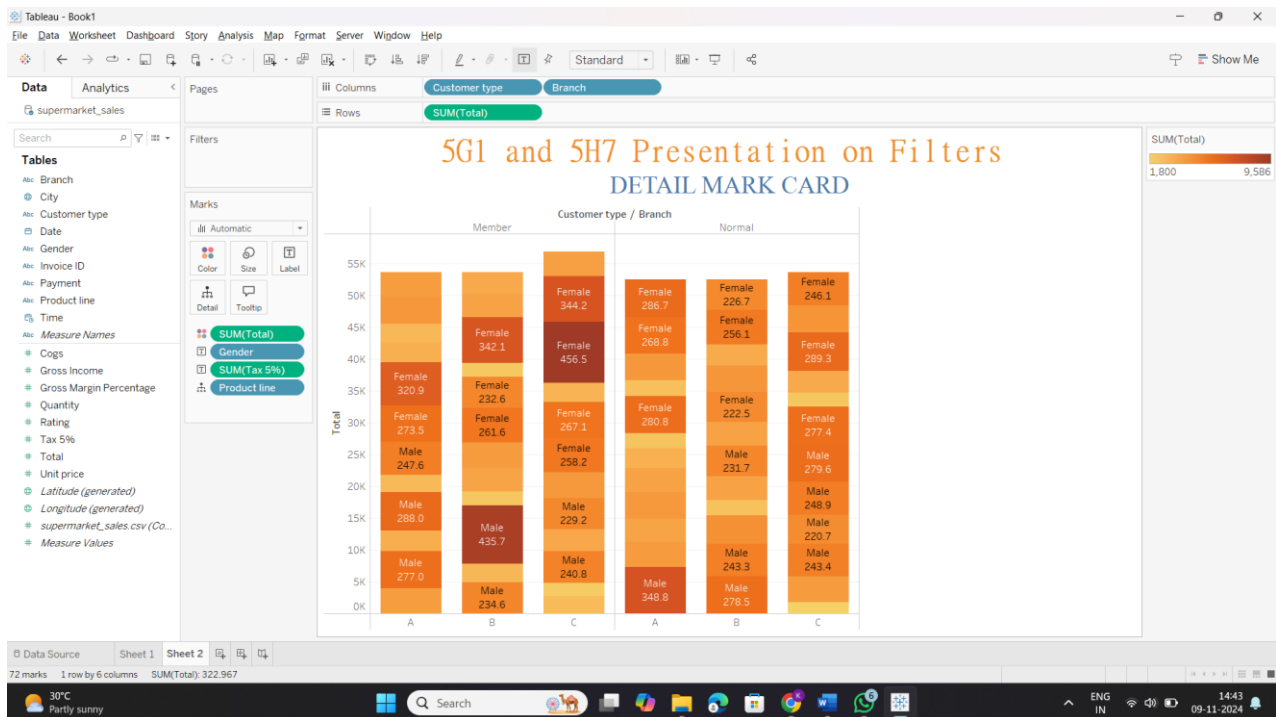
Step 04: Drag and drop the Gross Income field into tooltip marks card and after click any bar, automatically showing the profit in pop-up window.



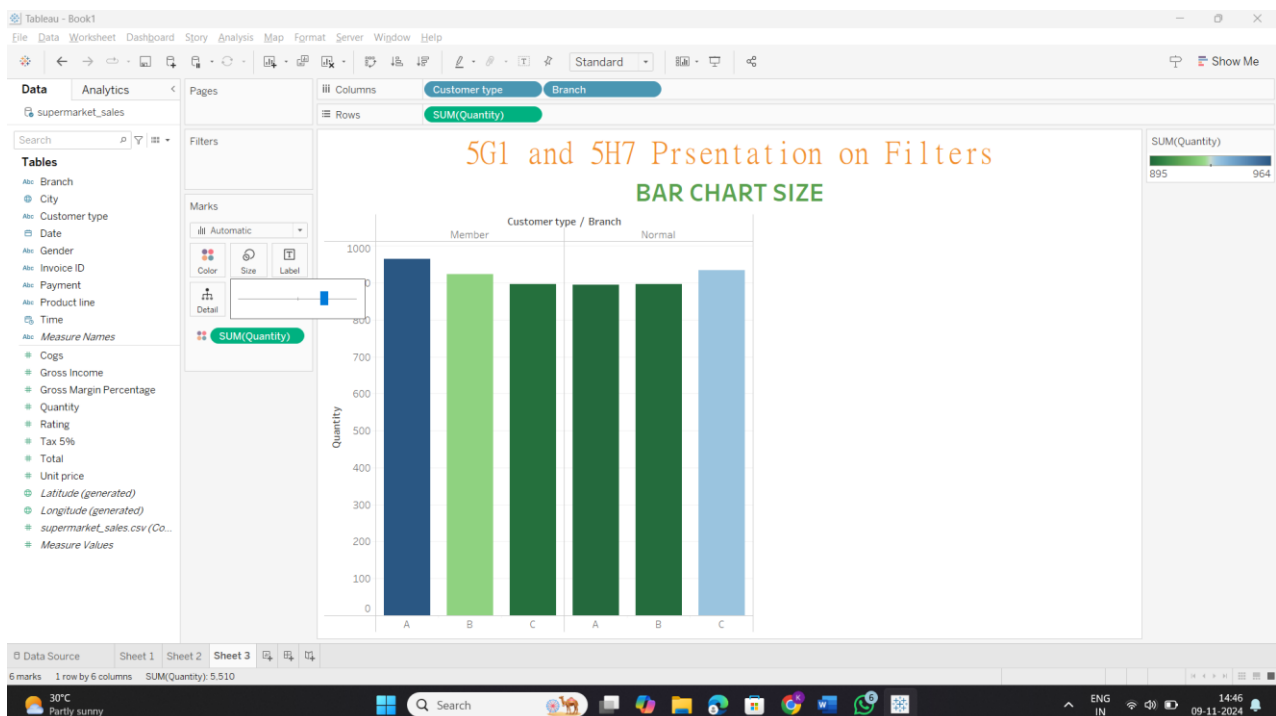
Step 05: Drag and drop the fields into row and columns self, automatically create bar chart. And drag and drop the Product Line field into Detail mark card. After drop, each bar is segmented into no of regions.



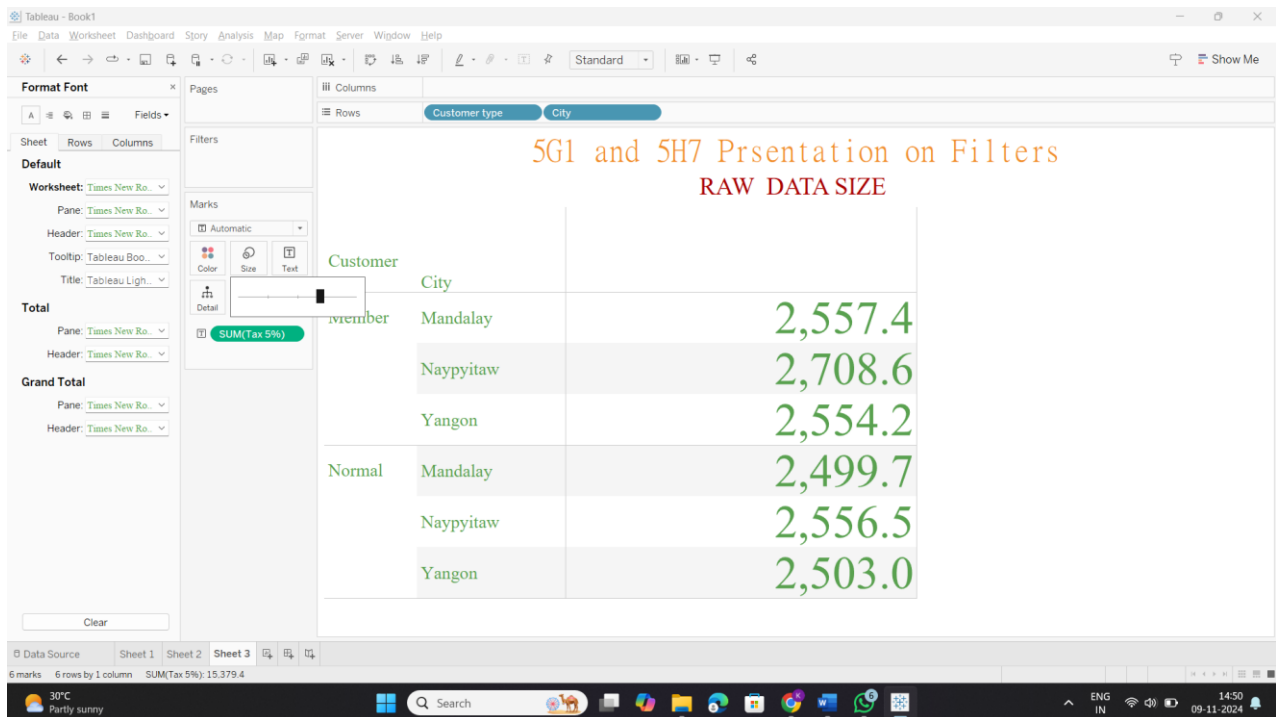
Step 06: Drag and drop the Tax, Product Line and Gender fields into Label Mark card.



Step 07: Drag and drop the Quantity and Customer type into row and columns marks, automatically created the bar chart and using size Mark card, increase the each bar width.



Step 08: Drag and drop the Quantity and City fields into row and columns marks and using size Mark card ,increase the row data (Numerical values) size.



CONCLUSION

In conclusion, embarking on the Tableau is a powerful tool for turning raw data into clear, insightful visualizations. By following the steps above—connecting to your data, creating visualizations, building interactive dashboards, and sharing your insights—you can deliver compelling, data-driven narratives that empower decision-making.

Following best practices for clarity, interactivity, and design will help ensure that your Tableau projects are impactful and effective. Whether you're analyzing business trends, research data, or performance metrics, Tableau provides the tools to bring your data to life.

By following the steps outlined in this guide, you can implement Tableau effectively in your project and deliver impactful visualizations that help stakeholders make informed decisions.

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